AN INDEPENDENT WORKBOOK DRAWING ON **THE COLLISION**



DESIGNING YOUR MIND FOR THE AI ERA

Are you using AI to outsource what makes you valuable?

Eric C. So
MIT PROFESSOR
AUTHOR OF **THE COLLISION: WHAT AI DOES TO US**

WORKBOOK (V1.1) · 6 PHASES
50–60 MINUTES · INDIVIDUAL AND PAIRED EXERCISES
PEN ONLY · NO AI

BEFORE YOU BEGIN · THE NEXT 50 TO 60 MINUTES

Overview

AI is increasingly capable. The quiet danger is handing off the thinking you're actually valued for to AI—until you can no longer do it as well yourself. The smartest way to use AI is to work backwards: start with the person you want to become and ask which thinking skills make that future version of you valuable.

THE FOUR RULES THAT GUIDE THE WORK

RULE 1

Value the struggle.

The struggle is what builds capability. Skipping it can deliver short-term gains at the cost of longer-term dependence on AI.

RULE 2

Value who you are without AI.

Who you are without AI is the set of things you can still do when the tool is taken away. A few signature thinking skills sit at the core of that—protect them.

RULE 3

Reinvest your surplus.

When AI gives back time, aim it at work that was out of reach before or at building new capabilities, not at more busywork or scrolling.

RULE 4

Make AI your cognitive trainer.

Use it to challenge your thinking, not perform it. Ask what it can teach you to do for yourself.

You need a pen, 50 to 60 minutes, and a partner for three of the six phases.

No AI during this exercise. The point is to think about your relationship with AI without letting it think for you.

1	**Your AI Use Snapshot** — *Ground your relationship with AI in recent evidence.*	5 MIN · SOLO	2
2	**Future-Self Portrait** — *Name the person you want to be, and the skills they depend on.*	6 MIN · SOLO	3
3	**Mirror with a Partner** — *Find the gap between what you wrote and what lands.*	5 MIN · PAIRS	4
4	**Map the Grid** — *Plot your tasks. Signature skill or not; complement or substitute.*	8 MIN · SOLO	5
5	**Defend Your Placements** — *Trade worksheets. Expect to move at least one.*	5 MIN · PAIRS	7
6	**Walk Through the Rules** — *Run one task through Rules 1, 3, and 4.*	18 MIN · PAIRS	8
—	**If You Have More Time** — *The private confession. Only if time permits.*	OPTIONAL	11
—	**Appendix: A Fully Worked Example** — *Every exercise, completed. Use when stuck.*	REFERENCE	12

PHASE ONE

1 Your AI Use Snapshot

5 MINUTES · INDIVIDUAL

Before deciding where AI belongs, start with evidence from your own life rather than an impression of it. What you can actually recall is the raw material for everything that follows: several of the uses that bear on your work, learning, or future self will reappear on your grid later.

WHAT TO LIST

List up to four times AI entered your work or life in the past seven days: what you were trying to do, and the role AI played. Include small, ordinary uses, not only the consequential ones. If you used AI little or not at all, include moments you considered it and chose not to.

Recent AI uses

Specific tasks, not general habits

WHAT I WAS TRYING TO DO	AI'S ROLE

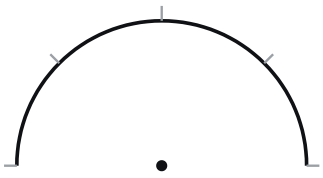
Now read the pattern

Use the examples above as evidence

Draw one needle on each gauge: start at the center dot and draw a line toward the point on the arc that best describes your overall pattern. If your examples pull in different directions, that is normal; mark where they land on balance. The task-by-task version comes in Phase 4. There is no ideal dashboard. A completed one, the author's own, sits on page 12.

AI Reach

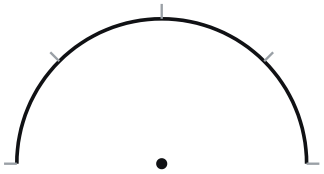
How much of my life does AI touch?



ONE ISOLATED CORNER — MANY PARTS OF MY LIFE

AI Gravity

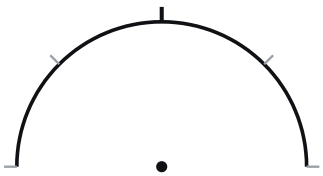
How strong is AI's pull on me — from convenience, deadlines, expectations, or fear of falling behind?



NO PULL — VERY STRONG PULL

Thinking Ownership

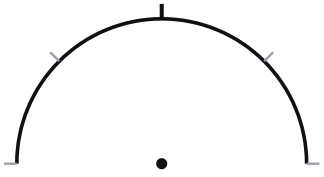
Across these examples, who did the core thinking?



AI DID MOST — I DID MOST

Capability Trajectory

If I kept using AI this way for a year, what would happen to my ability to do similar work without it?



LESS CAPABLE — MORE CAPABLE

2 PHASE TWO

Future-Self Portrait

6 MINUTES | INDIVIDUAL

Write a short profile of who you most want to be in five years, the way someone might describe you at your best. Three to five sentences. Be specific about what makes you valuable, not job titles or credentials. What thinking skills set this future version apart?

> **SIGNATURE SKILLS DESCRIBE HOW YOU THINK, NOT WHAT YOU DO OR KNOW**
> **Good examples:** "Synthesizing across disciplines," "spotting questions others have missed," "reading data with judgment," and "translating technical findings into plain English."
> **Bad examples:** "Knows Python," "MBA," and "Investment banker."

YOUR FUTURE-SELF PROFILE

2 PHASE TWO, CONTINUED

Name the Signature Skills

When you have finished the profile, go back and underline the three to five signature thinking skills future you depends on. Then write them out here, where the rest of the exercise can get at them.

Every field has one: the physician's diagnostic intuition, the lawyer's sense of how to craft persuasive arguments, the translator's feel for nuance, the analyst's nose for the number that doesn't belong. What are yours?

YOUR UNDERLINED SIGNATURE THINKING SKILLS

1.

2.

3.

4.

5.

3 PHASE THREE

Mirror with a Partner

5 MINUTES PAIRS

Discuss with the person(s) next to you. Each of you takes about two and a half minutes.

1. **Read your future-self portrait out loud.**
2. **Your partner gives one sentence back, beginning with the phrase:** "What I heard you say you want to be known for is..."
3. **Check it against the list above. Did your partner name one of the skills you just wrote down? If not, that gap is the finding.**
4. **Fix the phrase in your profile, then fix the list to match. If it landed, great. Switch roles.**

> WHY THE MIRROR MATTERS
>
> The point of the mirror is to catch the gap between what you think you wrote and what actually lands when a stranger reads it. Most people discover the first draft was vaguer than they realized.

4 PHASE FOUR

Map the Grid

8 MINUTES | INDIVIDUAL

Begin with any uses from Phase 1 that bear on your work, learning, or future self. Add other tasks you will encounter on the path to that future self until you have five to ten. On the 2×2 on the next page, place each task where it lands for you *right now*, not where you wish it landed.

If your Phase 1 record includes a moment when you considered or chose not to use AI, place that task where it would have landed if you had used it.

THE VERTICAL AXIS

Does this task involve one of your signature skills? Yes or no.

- **What "involves" means:** the test is one question. **If AI did this for a year, would I still be good at the thing it serves?** A task passes either because *the skill is in play* (the central task where that thinking happens, or work directly serving it), or because *the task feeds the skill* even when the skill is not in play. Drafting the argument: in play. Reading the underlying cases first: feeds it. The second kind is easy to miss because it looks rote.
- **The bottom row is everything else:** the tasks where neither is true. Not everything involves a signature skill, and if everything did, the term would lose its meaning. Bottom-row tasks can still require judgment and offer opportunities to learn; they simply do not involve one of your signature skills.

THE HORIZONTAL AXIS

Does AI complement your thinking, or substitute for it?

Complementing means AI contributes to the work while you do the core thinking. Substituting means AI does most of the core thinking and you mostly review or accept.

- **Complementing covers more than people expect.** AI can work *around* the thinking (clearing the path to it), *push on* it (arguing against your position, quizzing you), or *extend* its reach (work you could not have done alone). It only counts as substituting when AI does the thinking itself.

> **THIS IS AN AUDIT, NOT A VISIONING EXERCISE**
> Your placements will be challenged, and you should expect to be wrong about at least one of them. Your partner's job is to find it.

> **IF THE AXES STILL FEEL ABSTRACT**
> Page 13 shows a completed grid with twelve real tasks, placed and explained. One minute with it is usually enough to unstick a placement.

PHASE FOUR · WORKSHEET

Where Does AI Sit in Your Work?

INVOLVES A SIGNATURE SKILL?	Complements AI contributes to the work; I do the core thinking.	Substitutes AI does most of the core thinking; I mostly review or accept
YES	**Amplify** *grow what makes you you*	**Danger zone** *silent drift away from you*
NO	**Efficiency zone** *work with AI on non-signature tasks*	**Delegate zone** *hand non-signature tasks to AI*

FIVE TO TEN TASKS · WHERE THEY LAND TODAY, NOT WHERE YOU WISH THEY LANDED

PHASE FOUR, CONTINUED

4 Find Your Choice Tasks

Now look at the tasks in the top row, the ones that involve a signature skill. For each, ask: *"Am I keeping the thinking here, or drifting toward letting AI do it?"* Circle the ones that could honestly go either way depending on how you choose to use AI. Those are your **choice tasks**, and they are where the rules have the most bite for you.

NOW CHECK YOUR PHASE 1 GUESS

Look only at the top row, which contains the tasks that involve a signature skill. How many of those sit in the **Danger zone**? Set that against the Thinking Ownership needle you drew in Phase 1. Do the two tell a consistent story? **If they do not, note the mismatch and do not resolve it yet.** Bottom-row tasks do not count in this comparison.

WHAT THE GRID DOES NOT SEE

Your Reach needle covered your whole life; this grid covers only the work that bears on your future self. Name one thing AI touches that did not make it onto the grid, and why you left it off.

PHASE FIVE

5 Defend Your Placements

5 MINUTES PAIRS

Trade worksheets. Each of you picks two of your partner's placements and asks one of these questions about each.

- "Why this quadrant and not the one next to it?"
- "What makes you sure you are still doing the core thinking here, and AI is not quietly doing it for you?"
- "Is the rote work you handed off actually where you get your reps?"

Two minutes each way. Then a minute to revise any placements that did not survive the questions.

PLACEMENTS YOU MOVED, AND WHY

6 PHASE SIX

Walk Through the Rules

18 MINUTES PAIRS

First, the future-self portrait and the skills list you just built are **Rule 2 in action**—valuing who you are without AI. You have already done Rule 2, which is why the walkthrough below jumps to Rules 1, 3, and 4 and builds on what you named.

Pick one signature skill and a **Danger-zone** task that uses it; you will examine it under Rules 1 and 4: value the struggle, and make AI your cognitive trainer. If more than one qualifies, take the one where AI's pull on you is strongest, since that is the one most likely to slide. Rule 3, reinvest your surplus, looks at the other side of your grid: the **Efficiency- and Delegate-zone** tasks that free up the time you will put to work. Your partner probes each answer.

RULE ONE

Value the Struggle

Valuing the struggle is not about character. It is about what gets lost when the struggle is outsourced. AI writes your argument, but you cannot defend it. AI generates your presentation, but you stumble through the questions. Using AI on the work you are trying to learn from is like bringing a forklift to the gym. The struggle is the training.

> **THE QUESTION**
> Look at your Danger zone. Handing AI the thinking here is cognitive short-termism: you book the gain now—speed, ease, a finished deliverable—and suffer the loss later. **What specifically would that loss look like?**

PARTNER ASKS

Is there a way to keep AI in the loop that preserves the productive part of the struggle—the thinking that builds your judgment—while stripping out the busywork that doesn't?

PHASE SIX, CONTINUED

If Your Danger Zone Is Empty

TREAT IT AS A CLAIM TO TEST, NOT A CLEAN RESULT

An empty Danger zone is the most common self-flattering outcome here. Re-check each Amplify task: are you truly in the driver's seat, or is that where you wish you were? Then scan your Phase 1 record for a signature-skill task where AI supplied the core answer, draft, or analysis you used. If it is still empty, note that an empty zone is exactly what your Thinking Ownership and Capability Trajectory needles predict—three self-flattering answers agreeing is not corroboration. Pick the Amplify task where AI's pull is strongest and run Rule 1 on it as a pre-mortem.

NOTES AND REVISIONS

RULE THREE

Reinvest Your Surplus

Look at your Efficiency and Delegate zones. Pick a task where AI genuinely saves you time. Estimate the hours per week it gives back. The shift that matters: from *"How can I do this faster?"* to *"What becomes possible now?"* That is not only the hours you got back; it is also work AI now puts in range that was out of the question before.

YOUR REINVESTMENT PLAN

Name the specific signature-skill activities—or previously out-of-reach work—that those hours will go into. Practicing the kind of demanding work that builds new skills; tasks that compound everything else you do.

THE TASK AI HELPS WITH OR TAKES OVER

HOURS PER WEEK IT GIVES BACK

WHERE THOSE HOURS WILL GO

PARTNER ASKS

How will you make that realistic rather than aspirational?

PHASE SIX, CONTINUED

Rule 4: Make AI Your Cognitive Trainer

Name a recurring practice, not a one-time prompt, that makes AI strengthen your signature skill rather than perform it. Look back at your Capability Trajectory gauge: this practice should move you toward greater capability without AI over time. The shift is from asking what AI can do for you to asking what it can teach you to do for yourself: a thought partner that challenges your assumptions and a tutor that makes hard things accessible, not a ghostwriter.

> **A FEW EXAMPLES TO GET YOU STARTED**
>
> Before I commit to a position, I ask AI to argue against it.
>
> Before an exam, I ask AI to quiz me on the material I am least confident in, rather than just give me the answer.
>
> Before I trust my read of a chart, I describe what I see and ask AI where it sees something different.

NAME YOUR SPARRING PRACTICE

PARTNER ASKS

How would you know if it is actually working?

WHAT TO CARRY OUT OF THE ROOM

01 **You are not designing how to use AI in general.** You are designing how it fits around the person you are becoming.

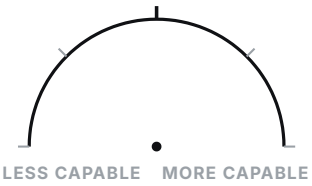
02 **Struggle is not always a problem to be solved.** It is often the investment that builds the skill.

03 **Reinvested time only counts if it builds something.** Not more email, not scrolling.

04 **A sparring practice is a pattern of use, not a clever prompt.** The habit is the point.

READ YOUR INSTRUMENTS AGAIN

You marked Capability Trajectory in Phase 1, before any of this. Mark it again now, and copy that first needle onto the same dial in a lighter hand. If the two differ, that gap is what you learned today.



THE ONE CHANGE YOU WILL MAKE THIS WEEK — AND HOW YOU WILL KNOW IT WORKED

OPTIONAL

If You Have More Time

ONLY IF TIME PERMITS

OPTIONAL PROMPT

It is five years from now. A peer got promoted ahead of you because they figured out how to use AI without losing what made them valuable. You did not.

In three to five sentences, write the private confession explaining what you got wrong. Ground it in something real—look back at your Phase 1 record and dashboard: a shortcut you have already taken, a source of AI Gravity you ignored, or a Capability Trajectory you allowed to continue. Name the specific task, not the habit in the abstract.

YOUR PRIVATE CONFESSION

APPENDIX

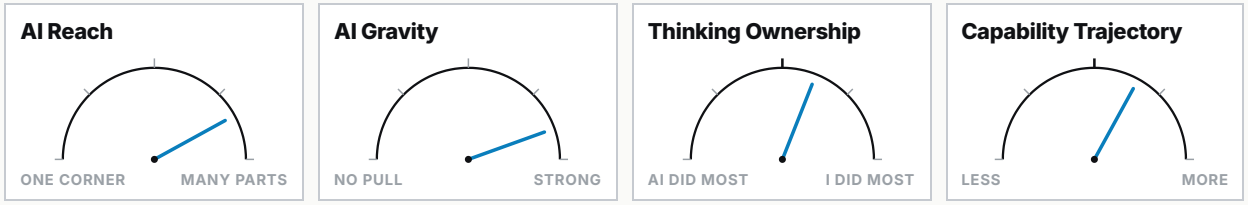
A Fully Worked Example

This appendix exists because the exercise can feel abstract until you watch someone do it with a real life. Read it before you start, or pick it up when you are stuck. The point is not to copy the answers but to see what a finished worksheet looks like.

> Eric C. So, an applied AI researcher at MIT Sloan who studies and teaches how AI is reshaping how people think and decide.

PHASE 1 · AI USE SNAPSHOT

WHAT I WAS TRYING TO DO	AI'S ROLE
Pressure-test a new framework before sharing it	Raised objections and alternative explanations; I evaluated them and revised the framework
Sort and triage a week of scheduling requests	Drafted the scheduling replies; I checked and sent
Design a new experiment from a vague idea	Proposed the design, and I used it largely as written



Three specific tasks, not "I use AI for research." All three of them turn up on my grid on page 13. I put Thinking Ownership right of centre—the third row is the one that makes that worth a second look in Phase 4.

PHASE 2 · FUTURE-SELF PORTRAIT

FUTURE-SELF PROFILE

Five years from now, I want to be recognized as a leading expert on what AI is doing to us. My research provides novel insights into how AI alters our perceptions and judgment in high-stakes settings. I am sought after as a teacher for how AI can be used to strengthen, rather than weaken, our cognitive abilities. My work speaks directly to practical questions and provides realistic frameworks that make it easier to thrive in the age of AI. I provide a voice of clarity that helps people make sense of conflicting bits of evidence and claims made by researchers and the media.

UNDERLINED SIGNATURE THINKING SKILLS — THE THREE TO FIVE THAT FUTURE ME REALLY DEPENDS ON

1. Designing studies that shed light on how AI changes our perception and judgment
2. Translating AI research into practical frameworks non-experts can use immediately
3. Pressure-testing claims about AI against evidence rather than vibes
4. Reading evidence with independence and sound judgment
5. Teaching how AI can be used to improve learning outcomes

APPENDIX · PHASE 4

Mapping the Grid

Here is a sample of tasks that come up in any given month. I am putting them on the grid where they have landed for me *recently*, not where I wish they landed.

THE RAW TASK LIST, BEFORE PLACEMENT

- Devising advice for a colleague who came to me for help
- Designing a new experiment from a vague idea
- Interpreting results from a study on AI use in high-stakes judgment
- Translating terminologies across cognitive science, ML, and behavioral finance
- Pressure-testing a new framework before sharing it with the field
- Creating a visual map of where ideas align versus conflict
- Editing prose for style after the ideas are settled
- Creating a specification document for an app or website design
- Writing code to implement tests I designed
- Sorting and triaging scheduling requests
- Creating images for presentations
- Formatting documents and filling out administrative forms

INVOLVES A SIGNATURE SKILL?	Complements AI contributes to the work; I do the core thinking.	Substitutes AI does most of the core thinking; I mostly review or accept
YES	**Amplify** *grow what makes you you* • Translating terminologies across cog sci, ML, and finance • Pressure-testing a new framework before sharing it • Visual map of where ideas align or conflict	**Danger zone** *silent drift away from you* ○ Devising advice for a colleague who came to me for help ○ Designing an experiment from a vague idea ○ Interpreting results from a new AI cognition study
NO	**Efficiency zone** *work with AI on non-signature tasks* • Editing prose for style after the ideas are settled • Creating a specification document for an app or website design • Writing code to implement tests I designed	**Delegate zone** *hand non-signature tasks to AI* • Creating images for presentations • Formatting documents and filling out administrative forms • Sorting and triaging scheduling requests

○ CHOICE TASKS — THEY COULD MOVE IF I CHANGED HOW I USED AI ON THEM

Notes on the Placements

WHY THE DANGER-ZONE TASKS ARE CIRCLED

I circled the three Danger-zone tasks because I could move them into the Amplify zone if I chose to use AI as a sparring partner rather than as a substitute. A common thread among these tasks is that what typically gets evaluated is the deliverable rather than the judgment behind it. Identifying that thread helps me identify related tasks where I might drift into the Danger zone before it happens.

WHY ROTE WORK CAN STILL SIT IN AMPLIFY

"Translating terminologies across fields" involves a signature skill even though the translating itself is rote, which is why it sits in Amplify rather than Delegate. AI handles the rote part, mapping similar or related ideas through the jargon of cognitive science, ML, and finance. This mapping is done in service of work that is critical for sharpening my signature thinking skills: getting to the core of the scholarship, asking the right questions, and making sense of competing ideas across papers. That is what Amplify means: AI helps me practice and refine the thinking skills I most value in myself.

DISTINGUISHING THE BOTTOM ROW

Three of the bottom-row placements are worth discussing because it is easy to get confused. These are tasks that matter but do not involve one of my signature thinking skills, and if everything counted as a signature skill the term would mean nothing. "Creating images for presentations" sits in Delegate: the images are useful, but I do not think of that kind of artistic work as part of what makes me valuable. "Creating a specification document for an app or website design" sits in Efficiency, not Delegate. Software and website design is not what defines my expertise as a professor, but the decisions about what to build are still mine. "Writing code to implement tests I designed" sits in Efficiency: I decide what to test and why; AI implements those decisions. The design is signature work, but coding is not. The "visual map" in Amplify and "creating images for presentations" in Delegate may look alike but sit at opposite corners. The reason is that the map helps me focus on the synthesis, which is central to my value, while the presentation image is decoration AI can produce end to end.

A NOTE FROM THE AUTHOR

This exercise, like the book behind it, is not anti-AI; it is about using AI to amplify your thinking rather than replace it. That includes the book itself. The stages that carry the thinking were mine to lead: forming the thesis, developing practical takeaways, and working out how the pieces fit together. They help build what I view as my signature thinking skills. I brought in AI as a writing coach to tailor the prose, which is why "editing prose for style" sits in my Efficiency zone and not my Danger zone.

> *"Using AI" was never yes or no for me. It was a question of where in the work it belonged.*

Walking One Choice Task Through the Rules

I am picking **"designing a new experiment from a vague idea."** It is the most consequential Danger-zone task on my list, and the one where substitution drift is most subtle.

RULE 1 · VALUE THE STRUGGLE

WHAT WOULD I LOSE IF AI DESIGNED THE EXPERIMENT FULLY?

The decisions about what to measure, what to hold constant, and which tests cleanly isolate the effect that is central to my hypotheses. AI offers to hand me a plausible design that resembles a hundred published ones but fails to speak to the intuition that motivated my hypotheses.

How I will keep AI in the loop but protect my signature skills: the productive struggle is in choosing the core hypothesis and the experimental design that isolates the effect. The unproductive part is the scaffolding around that productive struggle—explaining standard methods, formatting the protocol, chasing citations. I hand that to AI once the core decisions are mine.

THE PROTECTED STRUGGLE

Choose the core hypothesis and the experimental design that isolates the effect. Hand AI the scaffolding only after the core decisions are mine.

RULE 3 · REINVEST YOUR SURPLUS

WHERE THE RECLAIMED HOURS GO

I can reclaim time in either bottom zone. My Delegate-zone examples include scheduling requests, presentation images, and document formatting and forms. What I want to avoid is letting those hours get soaked up by more email, unnecessary meetings, or scrolling. My plan is to reinvest them into work that sharpens my signature skills: forming hypotheses and devising new ways to reach students.

Rule 4: Make AI Your Cognitive Trainer

MY SPARRING PRACTICE

I use AI as a sparring partner. I ask it to argue why something I proposed will not work for what I want to establish. Where are the confounds? What would a hostile referee attack first? What alternative explanation survives my controls? I read the critique, then decide deliberately which objections to act on rather than accepting or dismissing them wholesale. I sometimes run a parallel practice on tailoring prose: after I draft a section, I ask AI for competing versions of the same passage and coaching on the choices behind them. I question its suggestions and revise the passage myself. That back-and-forth helps the lessons stick.

HOW I WOULD KNOW IF IT IS WORKING

My study designs survive the review process with cleaner controls and sharper comparisons over time, and my writing stays true to my original thinking while reaching a mainstream audience. If my ability to do the core thinking without AI declines, I have drifted into the Danger zone and need to recalibrate.

Design the role of AI around the person you are becoming.

THE COLLISION

WHAT AI DOES TO US · ERIC C. SO